

3	(a) field strength = potential gradient correct sign OR directions discussed	M1 A1	[2]
	(b) area is $21.2 \text{ cm}^2 \pm 0.4 \text{ cm}^2$ (if outside $\pm 0.4 \text{ cm}^2$ but within $\pm 0.8 \text{ cm}^2$, allow 1 mark) 1.0 cm^2 represents $(1.0 \times 10^{-2} \times 2.5 \times 10^3 =)$ 25 V potential difference = 530 V	C2 C1 A1	[4]
	(c) $\frac{1}{2}mv^2 = qV$ $\frac{1}{2} \times 9.1 \times 10^{-31} \times v^2 = 1.6 \times 10^{-19} \times 530$ $v = 1.37 \times 10^7 \text{ ms}^{-1}$	C1 A1	[2]
	(d) (i) $d = 0$	B1	[1]
	(ii) acceleration decreases then increases some quantitative analysis (e.g. minimum at 4.0 cm) (any suggestion that acceleration becomes zero or that there is a deceleration scores 0/2)	B1 B1	[2]

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4. (c) rms output = $0.4/2$ or peak input = $220\sqrt{2}$ C1